作业14.

16.(1). 设曲面 $\vec{r}=(u, v, 2-u)$.

$\therefore \frac{\partial \vec{r}}{\partial u}=(1,0,-1)$, $\frac{\partial \vec{r}}{\partial v}=(0,1,0)$.

$\therefore \vec{r}_u \times \vec{r}_v=(1,0,1)$.

$\therefore$ 原式 $=\frac{\sqrt{2}}{2}\iint_{\Sigma}[P(x,y,z)+R(x,y,z)]dS$.

17.(2). 曲面 $\vec{r}=(R\sin\varphi\cos\theta, R\sin\varphi\sin\theta, R\cos\varphi)$.

$\therefore \vec{r}_\varphi=(R\cos\theta\cos\varphi, R\sin\theta\cos\varphi, -R\sin\varphi)$.

$\vec{r}_\theta=(-R\sin\varphi\sin\theta, R\sin\varphi\cos\theta, 0)$.

$\therefore$ 单位法向量 $\hat{n}=(\frac{x}{R}, \frac{y}{R}, \frac{z}{R})$. $\vec{F}=(0,0,x^2y^2z)$.

$\therefore$ 原式 $=\iint_{\Sigma}\frac{x^2y^2z^2}{R}dS$

$=\iint_{\Sigma}x^2y^2z^2R\sin\varphi\, d\varphi d\theta$.

$=\int_0^{2\pi}d\theta\int_{\frac{\pi}{3}}^{\pi}R^7\cos^2\theta\sin^2\theta\sin^5\varphi\cos^2\varphi\, d\varphi=\frac{2\pi}{105}R^7$.

(4). 单位法向量 $\hat{n}=(x,y,z)$. $\vec{F}=(x^2,y^2,z^2)$.

令 $x=\sin\theta\cos\varphi$, $y=\sin\theta\sin\varphi$, $z=\cos\theta$.

$\therefore dS=\sin\theta\, d\theta d\varphi$.

原式 $=\iint_{\Sigma}(x^3+y^3+z^3)dS$

$=\int_{\frac{\pi}{3}}^{\pi}d\varphi\int_0^{\frac{\pi}{2}}(\sin^3\theta\cos^3\varphi+\sin^3\theta\sin^3\varphi+\cos^3\theta)\sin\theta d\theta$.

$=\int_{\frac{\pi}{3}}^{\pi}d\varphi\int_0^{\frac{\pi}{2}}\cos^3\theta\sin\theta d\theta=\frac{\pi}{8}$

Date. Page.

(7). 令 $x=2\cos\theta$, $y=2\sin\theta$.

$\therefore \vec{F}=(0,-y,z+1)$, $\hat{n}=(\cos\theta,\sin\theta,0)$.

$x+z=2 \Rightarrow z=2-2\cos\theta$. $ds=2d\theta dz$.

$\therefore \iint \vec{F}\cdot\vec{n}=\int_0^{2-2\cos\theta}dz\int_0^{2\pi}(-4\sin^2\theta)d\theta=-8\pi$

(8). $\vec{F}=(y,x,z^2)$, $\vec{r}=(u\cos v,\ u\sin v,\ v)$.

$\vec{r}_u=(\cos v,\sin v,0)$; $\vec{r}_v=(-u\sin v,\ u\cos v,1)$.

$\therefore \vec{r}_u\times\vec{r}_v=(\sin v,-\cos v,u)$,

$\therefore \iint \vec{F}\cdot\vec{n}=\iint_D u(-\cos 2v+v^2)\,du\,dv$

$=\int_0^{\pi}(v^2-\cos 2v)\,dv\int_0^1 u\,du=\frac{1}{6}\pi^3$

18.(1). $\vec{n}=(x,y,z)$.

令 $x=\sin\theta\cos\varphi$, $y=\sin\theta\sin\varphi$, $z=\cos\theta$

$\therefore 0\le\theta\le\frac{\pi}{2}$, $0\le\varphi\le\frac{\pi}{2}$, $dS=\sin\theta\,d\theta\,d\varphi$

$\therefore Q=\int_0^{\frac{\pi}{2}}d\theta\int_0^{\frac{\pi}{2}}(\sin^3\theta\cos^3\varphi+\sin^3\theta\sin^3\varphi+\cos^3\theta)\,d\varphi$.

$=\frac{3}{8}\pi$,

19.(1). $\begin{cases} dx=-3a\cos^2t\sin t\,dt \\ dy=3a\sin^2t\cos t\,dt. \end{cases}$

$\therefore A=\frac{1}{2}\oint_C(x\,dy-y\,dx)$

$=\frac{1}{2}\int_0^{2\pi}3a^2\sin^2t\cos^2t\,dt=\frac{3}{8}\pi a^2$.

(3). $\begin{cases} dx = a(1-\cos t)dt. \\ dy = a\sin t\, dt. \end{cases}$

$A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$

$= \frac{1}{2}\int_0^{2\pi}\left[a^2\sin t(t-\sin t) - a^2(1-\cos t)^2\right]dt$

$= -3\pi a^2 \quad \therefore S = 3\pi a^2.$

20. (4). $P = 2x\sin y - 4y,\ Q = x^2\cos y + x.$

$\partial_x Q = 2x\cos y + 1,\ \partial_y P = 2x\cos y - 4$

原式 $= \iint_D (\partial_x Q - \partial_y P)\,dA = 5\pi \times 3 = 15\pi$

(3). $P = x^2 y - 2y,\ Q = \frac{x^3}{3} - x.$

$\partial_x Q = x^2 - 1,\ \partial_y P = x^2 - 2.\ \partial_x Q - \partial_y P = 1.$

$\therefore$ 原式 $= \frac{1}{2}$

(5). 积分: $(1,0) \to (\pi+1, 0).$

$\therefore P = \sqrt{x^2+y^2},\ Q = y(xy + \ln(x + \sqrt{x^2+y^2})).$

$\therefore \partial_x Q - \partial_y P = y^2.$

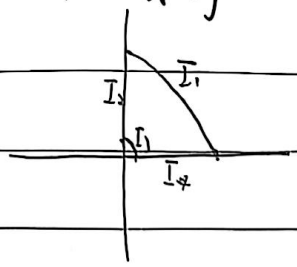
$\therefore \iint_D (\partial_x Q - \partial_y P)\,dA = \int_1^{\pi+1} dx \int_0^{\sin(x-1)} y^2\,dy = \frac{4}{9}.$

$\int_{x=1}^{\pi+1} P\,dx = \frac{\pi^2}{2} + \pi$

$\therefore$ 原式 $= \frac{4}{9} - \frac{\pi^2}{2} - \pi$

Date. Page.

(7). $P=\frac{y}{x^2+y^2}$, $Q=-\frac{x}{x^2+y^2}$.



如左图构造回路

$\therefore \partial_x Q-\partial_y P=0$.

$\therefore I_2=I_4=0$.

$I_3=\frac{\pi}{2}$ $\quad\therefore I_1+I_3=0$

$\therefore$ 原式 $=I_1=-\frac{\pi}{2}$

2). (1). $P=x+y$, $Q=x-y$.

$\therefore \frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}=1$ $\quad\therefore$ 路径无关.

势函数 ϕ: $\frac{\partial\phi}{\partial x}=P$; $\frac{\partial\phi}{\partial y}=Q$.

$\therefore \phi=\frac{1}{2}x^2+xy-\frac{1}{2}y^2$.

$\therefore$ 原式 $=\phi(2,2)-\phi(1,0)=\frac{7}{2}$

(3). $P=(y+e^{-x}\sin y)$, $Q=x-e^{-x}\cos y$.

$\frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}=1+e^{-x}\cos y$. $\quad\therefore$ 路径无关.

势函数 ϕ: $\frac{\partial\phi}{\partial x}=P$, $\frac{\partial\phi}{\partial y}=Q$

$\therefore \phi(x,y)=xy-e^{-x}\sin y$.

$\therefore$ 原式 $=\phi(1,\frac{\pi}{2})-\phi(0,0)=\frac{\pi}{2}-\frac{1}{e}$

Date. Page.

22.(2). $P = 1 - \frac{y^2}{x^2}\cos\frac{y}{x}$, $Q = \sin\frac{y}{x} + \frac{y}{x}\cos\frac{y}{x}$.

$\therefore \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = -\frac{2y}{x^2}\cos\frac{y}{x} + \frac{y^2}{x^3}\sin\frac{y}{x}$

可: $\frac{\partial u}{\partial x} = P$, $\frac{\partial u}{\partial y} = Q$

$\therefore u(x,y) = x + y\sin\frac{y}{x}$.

23.(2). $P = f(x)y$, $Q = -f(x)x$.

$\therefore f'(x)x + f(x) = -f(x)$.

$\therefore 2f(x) + xf'(x) = 0$.

$\therefore \ln(f(x)) = -2\ln x + C$

$\because f(1) = 1$ $\therefore C = 0$.

$\therefore f(x) = x^{-2}$.

24. $P = 2xy$, $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = 2x$.

$\therefore Q(x,y) = x^2 + h(y)$.

势函数: $\Phi(x,y) = x^2y + H(y)$, $\frac{\partial \Phi}{\partial y} = x^2 + H'(y) \Rightarrow H'(y) = h(y)$.

可: $\Phi(t,1) = \Phi(1,t)$.

$\therefore t^2 + H(1) = t + H(t)$.

$\therefore h(t) = H'(t) = 2t - 1$.

$\therefore Q(x,y) = x^2 + 2y - 1$.

Date. Page.

25. $P=\frac{x^y}{y^2}(x^2+y^2)^p$, $Q=-\frac{x^2}{y^2}(x^2+y^2)^p$.

$\therefore \frac{\partial P}{\partial y}=\frac{\partial Q}{\partial x} \Rightarrow p=-\frac{1}{2}$

即: $P=\frac{x}{y\sqrt{x^2+y^2}}$, $Q=-\frac{x^2}{y^2\sqrt{x^2+y^2}}$.

势函数: Φ : $\begin{cases}\frac{\partial \Phi}{\partial x}=P & ① \\ \frac{\partial \Phi}{\partial y}=Q & ②\end{cases}$

①: $\Phi=\frac{1}{y}\sqrt{x^2+y^2}+h(y)$.

$\therefore \frac{\partial \Phi}{\partial y}=-\frac{x^2}{y^2\sqrt{x^2+y^2}}+h'(y)=Q \quad \therefore h(y)=0$.

$\therefore \Phi=\frac{1}{y}\sqrt{x^2+y^2}$.

$\therefore$ 原式 $=\Phi(0,2)-\Phi(1,1)=1-\sqrt{2}$.

26. (2). $P=(1+y)\cos xy$, $Q=x\cos xy$.

$\therefore \frac{\partial Q}{\partial x}=\cos xy+xy(-\sin xy)$

$\frac{\partial P}{\partial y}=\cos xy+xy(-\sin xy)$

$\therefore \frac{\partial Q}{\partial x}=\frac{\partial P}{\partial y}$.

设 $du=Pdx+Qdy$.

$\therefore \frac{\partial u}{\partial x}=P \Rightarrow u=x+\sin xy+h(y)$

$\therefore \frac{\partial u}{\partial y}=x\cos xy+h'(y)=x\cos xy \quad \therefore h(y)=C$

通解: $x+\sin xy=C$.

4) 两边同乘 $\frac{1}{x^2y^2}$

$\therefore \left(\frac{1}{x^2y}+\frac{2}{x}\right)dx+\left(\frac{1}{xy^2}-\frac{2}{y}\right)dy=0$

$\therefore M=\frac{1}{x^2y}+\frac{2}{x},\quad N=\frac{1}{xy^2}-\frac{2}{y}$

$\therefore \frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}=-\frac{1}{x^2y^2}$

设 $du=Mdx+Ndy$.

$\therefore \frac{\partial u}{\partial x}=M \Rightarrow u=-\frac{1}{xy}+2\ln|x|+h(y)$

$\because \frac{\partial u}{\partial y}=N \therefore \frac{1}{xy^2}+h'(y)=\frac{1}{xy^2}-\frac{2}{y}$

$\therefore h'(y)=-\frac{2}{y} \quad \therefore h(y)=-2\ln|y|+C$.

$\therefore$ 解: $-\frac{1}{xy}+2\ln|x|-2\ln|y|=C$

Date. Page.

思考题：

(1)、先证：充分性. $\Delta u=0$ 时

$$\oint_{\partial S}\frac{\partial u}{\partial n}ds=\iint_S \Delta u\,dxdy=0.$$

再证：必要性，$\oint_{\partial S}\frac{\partial u}{\partial n}ds=0$

取定：$P\in S$. 以P为圆心，ε为半径作圆弧 S_ε.

$$\therefore \iint_{S_\varepsilon}\Delta u\,dxdy=\oint_{\partial S_\varepsilon}\frac{\partial u}{\partial n}ds=0.$$

令 $\varepsilon\to 0$ $\therefore \Delta u\cdot\pi\varepsilon^2=0$ $\therefore \Delta u(P)=0.$

即 $\Delta u=0$.

(2)、取向量场：$\vec{F}=u\nabla u$.

$\therefore \nabla\cdot(u\nabla u)=(\nabla u)\cdot(\nabla u)+u\Delta u$.

$$\therefore \iint_S[(\nabla u)^2+u\Delta u]\,dxdy=\oint_{\partial S}(u\nabla u)\cdot n\,ds$$

$$=\oint_{\partial S}u\frac{\partial u}{\partial n}ds.$$

$$\therefore 即：\iint_S(\nabla u\cdot\nabla u)\,dxdy=\iint_S\left[\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2\right]dxdy$$

$$=-\iint_S u\Delta u\,dxdy+\oint_{\partial S}u\frac{\partial u}{\partial n}ds.$$

(3)、证：u, v 均为调和函数，$u|_{\partial S}=v|_{\partial S}\Rightarrow u=v$

令 $w=u-v$.

$\therefore \Delta w=0$, $w|_{\partial S}=0$.

$$\therefore \iint_S(\nabla w)^2\,dxdy=-\iint_S w\Delta w\,dxdy+\oint_{\partial S}w\frac{\partial w}{\partial n}ds.$$

$$=0.$$

$\therefore \nabla w=0$ 又 $\because w|_{\partial S}=0$ $\therefore w=0$ 即 $u=v$